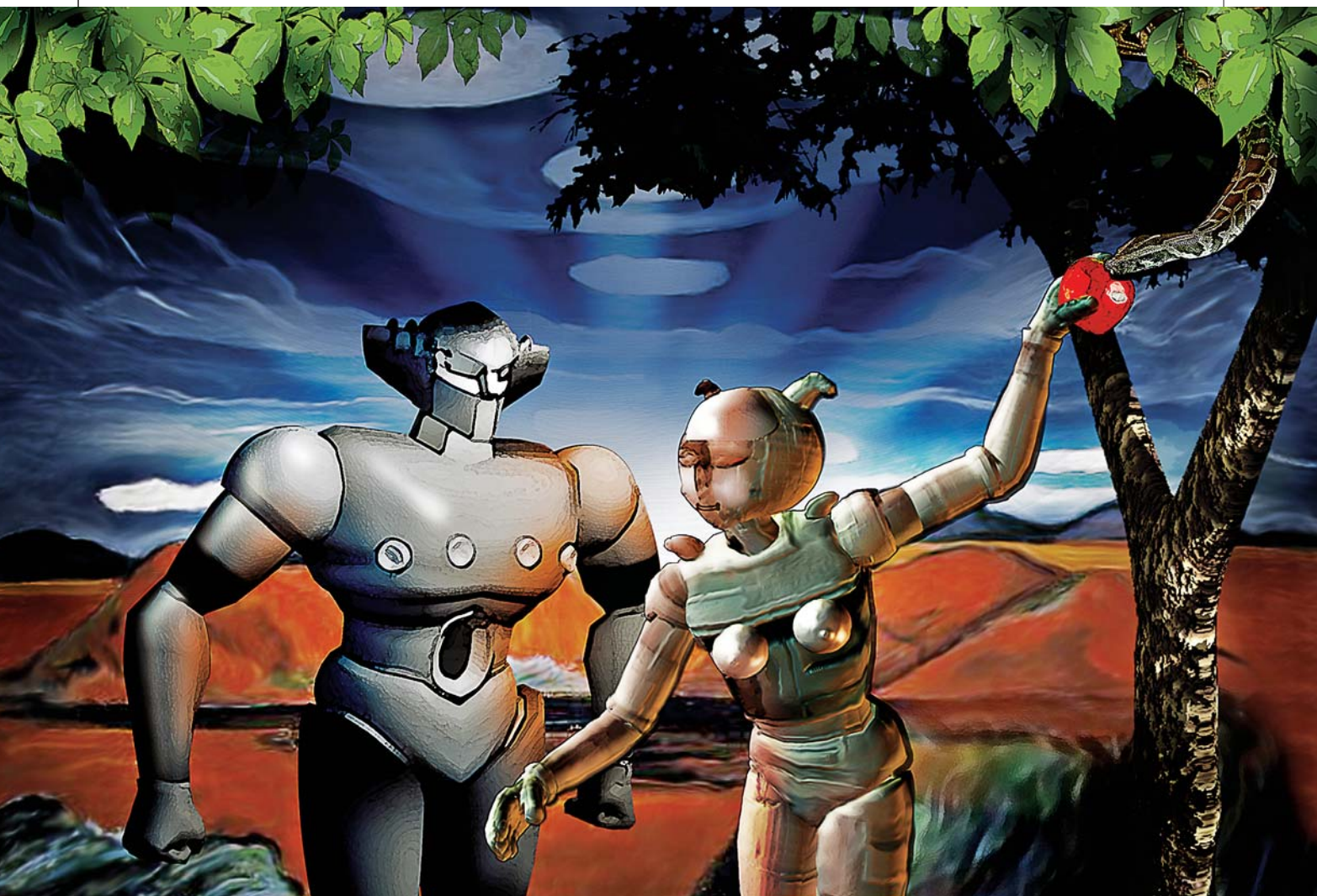




Imaging Sani Mani

THE COMRADES AND THEIR Fantastic Robots

Four Bangalore engineers try Genesis of a different kind—they have built a robot couple they like to call Adam and Eve. Evolution is not far behind

Suprotip Ghosh

It was mostly made from junk! I remember we actually used a hair clip as a gripper to start with,” recalls Jnaneshwar Das, as he ruminates upon his college days, back to the origins of COMRADE—the Co-Operative Mobile Robots for Autonomous Decisive Exploration project.

Not a multi-billion dollar, deep-space-exploration-probe this, but a university project that four students undertook at the Peoples Education Society Institute of Technology (PESIT), Bangalore. The result was unique in its application

and sheer innovation. Adam and Eve, COMRADEs both, headed the same way as another such endeavour half a world away: the Mars Rover project, started and developed by NASA’s Jet Propulsion Laboratory is similar in philosophy and design to COMRADE. Besides, the application of Open Source software for Rover was a first for any NASA project.

COMRADE involves construction of “networked, autonomous mobile robots, equipped with sensors, grippers and onboard computers.” Networking is achieved through distributed task resolution via a wireless link: take a bunch of robots and give them a task in the form of a